



OPEN Accurate AQI forecasting in a high-altitude city using a simulated CVOCA-BiLSTM hybrid model: a case study of Lhasa, Tibet

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Forecasting of the Air Quality Index (AQI) in high-altitude cities is significantly challenged by nonlinear temporal dynamics, particularly due to complex interactions between atmospheric physics and pollutant transport. A hybrid forecasting model integrating a simulated Complex-Valued Optical Convolution Accelerator (CVOCA) with a Bidirectional Long Short-Term Memory (BiLSTM) network is presented to improve prediction accuracy and computational efficiency. The simulated CVOCA module enables high-speed, complex-valued convolution to extract amplitude and phase features from high-dimensional AQI sequences, while the BiLSTM component captures bidirectional temporal dependencies. Performance was evaluated using daily AQI data from Lhasa (2014–2024, 4018 samples). The hybrid model achieved RMSE=1.60, MAE=0.93, MAPE=1.60%, R=0.997, and R²=0.996, outperforming standalone LSTM and BiLSTM models. The proposed model captures complex AQI dynamics in plateau environments and provides a scalable solution for real-time forecasting and environmental decision support.

Keywords AQI forecasting, CVOCA-BiLSTM hybrid model, Exceptional predictive performance, Tibet

Abbreviations

AdaBoost	Adaptive Boosting
ANN	Artificial Neural Network
AQI	Air Quality Index
ANFIS	Adaptive Neuro-Fuzzy Inference System
ARMA	Autoregressive Moving Average
BPTT	Backpropagation Through Time
BR	Bayesian Ridge
BWO	Beluga Whale Optimization
CAformer	Convolutional Block Attention Module augmented Informer
CatBoost	Categorical Boosting
CNN	Convolutional Neural Network
CVOCA	Complex-Valued Optical Convolutional Accelerator
DTR	Decision Tree Regression
FCM	Fuzzy C-Means
FFT	Fast Fourier Transform
FNN	Feedforward Neural Network
GB	Gradient Boosting
GBDT	Gradient Boosted Decision Trees
GML	Geographics Machine Learning
GRU	Gated Recurrent Units
KNN	K-Nearest Neighbors
KNR	K-Neighbors Regressor
LARS	Least Angle Regression

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LarsCV	Least Angle Regression with Cross-Validation
LASSO	Least Absolute Shrinkage and Selection Operator
LightGBM	Light Gradient Boosting Machine
LSTM	Long Short-Term Memory
LR	Logistic Regression
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MHBA	Modified Honey Badger Algorithm
MLP	Multi-Layer Perceptron
OA	Orchard Algorithm
PBLH	Planetary Boundary-Layer Height
PSO	Particle Swarm Optimization
RF	Random Forest
RMSE	Root Mean Square Error
RNN	Recurrent Neural Networks
SLNet	Spatial-and-Local-Aware Deep Learning Network
STL	Seasonal-Trend Decomposition based on Loess
SVR	Support Vector Regression
TimeGAN	Time Generative Adversarial Networks
WKNN	Weighted K-Nearest Neighbor
XGBoost	Extreme Gradient Boosting

In recent years, the escalating frequency and intensity of large-scale wildfires worldwide have become a critical driver of atmospheric pollution, severely undermining urban environmental sustainability and endangering public health^{1,2}. Exceedance of hazardous Air Quality Index (AQI) thresholds constitutes a significant threat to ecosystem stability³.

The Tibetan Plateau, often called the “Third Pole” performs an indispensable function in both regional and global sustainability. Nonetheless, polar and high-altitude regions constitute sensitive interfaces within the Earth’s climate system. Residents of these regions face heightened risks from amplified climate variability and increased pollutant transport⁴. The Tibetan Plateau’s highly variable topography causes dramatic swings in Planetary Boundary-Layer Height (PBLH), directly affecting near-surface pollutant dispersion and deposition. Studies have indicated that warming over the Tibetan Plateau can enhance vertical mixing and may influence secondary aerosol formation processes, thereby contributing to increased complexity and variability in AQI time series⁵. Compared with urban plains, AQI monitoring stations across the Tibetan Plateau are sparse and unevenly distributed, yielding input data with poor spatial continuity and representativeness⁶. Moreover, the region’s dry climate and limited pollution sources produce relatively small AQI fluctuations and overly smooth time series, reducing model sensitivity and robustness to subtle changes⁷. For instance, over the past decade, Lhasa’s AQI exhibited a relative standard deviation (RSD) of 30.86% and a standard deviation (SD) of 17.81, which are substantially lower than those of Beijing (RSD: 65.09%, SD: 60.34) and Shanghai (RSD: 45.32%, SD: 33.22), highlighting the markedly weaker variability in Lhasa’s air quality data. Finally, the proportions of secondary aerosols versus primary emissions vary sharply with humidity and temperature, creating complex spatiotemporal interactions that conventional models struggle to capture⁸. Collectively, these factors—including the plateau’s unique high-altitude boundary-layer dynamics, sparse and smooth monitoring data, and complex transboundary transport—substantially elevate the challenge of AQI forecasting on the Tibetan Plateau.

AQI forecasting methods are generally classified into two primary categories: process-driven models and data-driven models⁹. Process-driven models are based on the principles of atmospheric physics and chemical mechanisms. These models simulate the generation, dispersion, transport, and deposition of pollutants to predict air quality. Representative examples include chemical transport models such as the Community Multiscale Air Quality (CMAQ) model and the Weather Research and Forecasting model coupled with chemistry (WRF-Chem)^{10,11}. These models require numerous input parameters, including meteorological fields, emission inventories, and topographic data. While process-driven models provide strong interpretability and are grounded in physical principles, they are computationally demanding and highly sensitive to the quality of initial input data, making their predictive accuracy susceptible to various uncertainties. In contrast, data-driven models focus on learning statistical patterns and latent relationships from historical datasets, without explicitly modeling the physical transport mechanisms of pollutants. This category encompasses traditional statistical methods (e.g., ARMA, SVR), machine learning algorithms (e.g., Random Forest, XGBoost), and deep learning architectures (e.g., LSTM, CNN, Transformer). Data-driven models are typically more flexible and capable of achieving higher predictive accuracy, particularly when modeling nonlinear and highly dynamic AQI patterns^{12–14}. However, pure temporal deep learning models without feature enhancement tend to overfit “flat” patterns when data is sparse and low in noise, while also incurring high computational costs that hinder real-time forecasting¹⁵. Ensemble machine learning algorithms offer limited threshold segmentation for smoothed AQI sequences and lack adequate modeling of temporal dependencies¹⁶. Therefore, recent studies on AQI and air pollutant forecasting have demonstrated a growing interest in hybrid models, which integrate feature engineering or decomposition techniques with advanced neural networks. These hybrid models have shown superior forecasting performance and robustness across diverse environmental contexts.

However, traditional hybrid models typically combine signal decomposition techniques—such as wavelet transform, Variational Mode Decomposition (VMD), or Empirical Mode Decomposition (EMD)—with unidirectional models like LSTM or SVR to process original real-valued AQI time series^{17,18}. While these methods have demonstrated effectiveness in capturing general trends, they often fall short in responding to

nonlinear fluctuations and abrupt shifts. Moreover, their performance is highly contingent upon the selection of decomposition parameters, thereby constraining their generalizability across diverse datasets and geographic regions. To effectively extract deep dynamic patterns from time series data, this study introduces a novel simulated Complex-Valued Optical Convolution Accelerator (CVOCA), built upon a complex-valued neural network architecture that preserves data integrity while enhancing phase and amplitude information capture, thereby effectively enhance the information of minor fluctuations and achieve richer hidden representations¹⁹. Subsequently, the extracted features are fed into a Bidirectional Long Short-Term Memory (BiLSTM) network for deep temporal modeling of both forward and backward dependencies. Moreover, regularization strategies combined with a robust loss function mitigated overfitting to flat patterns. The use of Huber loss, dropout, and early stopping ensured that training and validation curves converged smoothly, demonstrating strong generalization rather than mechanical memorization of smoothed segments. Collectively, the simulated CVOCA-BiLSTM framework significantly enhances the model's capability to fit complex temporal structures. With respect to predictive accuracy, robustness, and computational efficiency, the proposed model outperforms conventional hybrid approaches and exhibits significant potential for practical deployment.

This study proposes a hybrid deep learning model that integrates a simulated CVOCA with a Bidirectional LSTM to improve AQI forecasting in high-altitude cities. By leveraging simulated CVOCA's nonlinear feature extraction and BiLSTM's temporal dependency modeling, the framework offers improved accuracy and reliable performance under complex climatic conditions.

Materials and methods

Study area and data collection

The proposed simulated CVOCA-BiLSTM hybrid model was validated using AQI time series data from Lhasa, a representative high-altitude city on the Qinghai-Tibet Plateau. As shown in Fig. 1, AQI values were calculated as the mean of six monitoring sites—Xizang University, Railway Station, Monitoring Station, Barkhor Street, Environmental Protection Agency, and Radiation Station—using historical data from the China National Environmental Monitoring centre (<https://www.cnemc.cn/>) and the AQI Study database (<https://www.aqistudy.cn/historydata/>). The dataset comprised daily AQI values for Lhasa, Shanghai, and Beijing from January 1, 2014, through December 31, 2024, yielding a total of 12,054 observations. To ensure data integrity, the series was first examined for missing entries, and any gaps were filled using linear interpolation based on adjacent values. To improve training stability, all AQI inputs were normalized to the [0, 1] interval via MinMaxScaler, and model outputs were subsequently denormalized. Following preprocessing, the dataset was partitioned into training (70%), validation (15%), and testing (15%) subsets. The convolutional layer in the simulated CVOCA model was configured with 64 filters, a kernel size of 3, and the ReLU activation function to extract localized temporal features from the AQI time series. The BiLSTM component comprised three stacked layers with

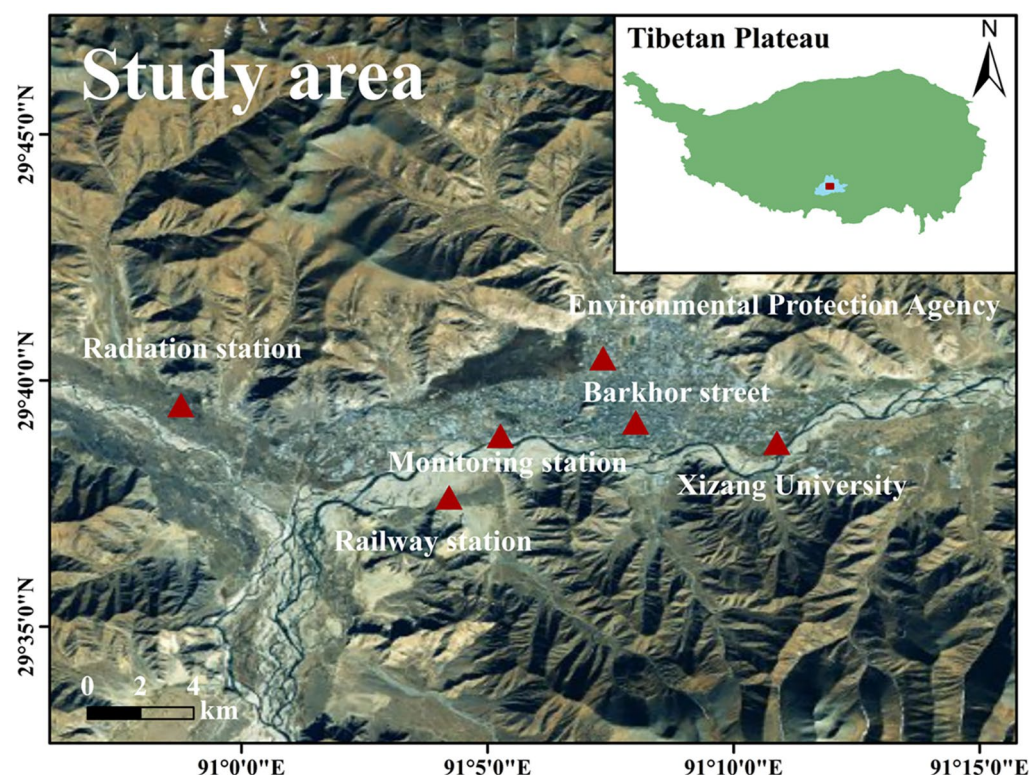


Fig. 1. Study area and air monitoring station location. Photos taken from Google Maps. Map data ©2025 Google.

256 hidden units each and a dropout rate of 0.1, using a time step of 10 and the tanh activation function to model temporal dependencies in both forward and backward directions. The model was trained for 100 epochs with a learning rate of 0.0005, utilizing CPU-based parallel computation to enhance efficiency. The optimized network architectures and hyperparameters of all models are summarized in Table S1 (Supporting Information). Model performance was assessed using five standard metrics: root mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), Pearson correlation coefficient (R), and coefficient of determination (R^2). The selection of Lhasa as a case study enables a robust evaluation of the proposed model's effectiveness and stability under high-altitude environmental conditions.

Methods

The methodology employed in this study involves the integration of a simulated CVOCA and a BiLSTM neural network. A brief overview of these methods is provided below:

Complex-Valued Optical Convolutional Accelerator (CVOCA)

The Complex-Valued Optical Convolution Accelerator (CVOCA) was first introduced by Bai et al.¹⁹ CVOCA is a novel optical convolutional framework designed for the extraction of high-dimensional, ultrafast, and hierarchical features from waveform or complex-valued data streams. In this study, simulated CVOCA—not only denoting a software-based algorithmic implementation of the complex-valued convolution operation but also encompassing specific simulations of amplitude and phase information in optical physical processes—is employed as a front-end feature extractor for AQI data, enabling the efficient decomposition of complex temporal patterns. The feature extraction process is mathematically represented in Eqs. (1) and (2).

For a complex-valued input $X = |X|e^{j\varphi} = X_R + jX_I$, and a complex-valued convolutional kernel $W = |W|e^{j\theta} = W_R + jW_I$, the resulting complex-valued convolution can be expressed as:

$$Y[n] = \sum_{m=1}^M |W[m]| \cdot |X[n-m+1]| \cdot e^{j(\theta[m] + \varphi[n-m+1])} \quad (1)$$

Expanding this expression, the complex convolution result becomes:

$$Y[n] = (W_R * X_R - W_I * X_I)[n] + j(W_R * X_I + W_I * X_R)[n] \quad (2)$$

Here, $W_R[m] = |W[m]| \cdot \cos\theta[m]$; $W_I[m] = |W[m]| \cdot \sin\theta[m]$; $X_R[n] = |X[n]| \cdot \cos\varphi[n]$; $X_I[n] = |X[n]| \cdot \sin\varphi[n]$. X_R and X_I represent the real and imaginary parts of the input data, respectively. W_R and W_I represent the real and imaginary parts of the convolution kernel. * denotes the convolution operation.

Long Short-Term Memory (LSTM)

The Long Short-Term Memory (LSTM) network, introduced by Hochreiter et al.²⁰ incorporates an improved strategy to optimize the training of traditional Recurrent Neural Networks (RNNs). By using Backpropagation Through Time (BPTT), LSTM effectively mitigates the vanishing gradient problem³⁵. This model is particularly well-suited for handling and predicting significant events that occur with long time intervals or delays in time series data. Unlike conventional neurons, LSTM networks are composed of memory blocks connected across layers. Each LSTM layer consists of a set of recurrently connected memory blocks, which can be seen as differentiable versions of memory chips in digital computers.

The LSTM unit is controlled by three gating mechanisms: the forget gate, the input gate, and the output gate, along with a cell state. The mathematical expressions for these operations are given by Eqs. (3)–(8) as follows:

1. The forget gate f_t :

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (3)$$

Here, σ represents the sigmoid activation function, h_{t-1} is the hidden state at the previous time step, x_t is the current input, W_f and b_f are the weight and bias terms, respectively.

2. The input gate i_t and Candidate Cell State $\tilde{C}_t$:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (4)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (5)$$

3. Cell State Update C_t :

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (6)$$

Here, C_{t-1} represents the cell state from the previous time step.

4. Output Gate o_t and Final Hidden State h_t :

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (7)$$

$$h_t = o_t \cdot \tanh(C_t) \quad (8)$$

During the training process, the various gating mechanisms collaboratively contribute to the updating of weights. This process is visually depicted in the LSTM structure diagram shown in Fig. 2. It is important to note that the LSTM structure incorporates the Sigmoid activation function (σ) and the Tanh activation function, as well as element-wise addition ($\oplus$) and multiplication ($\otimes$) operations.

Bidirectional Long Short-Term Memory (BiLSTM)

Bidirectional Long Short-Term Memory (BiLSTM) was first introduced by Graves et al.²¹ as an enhanced version of the LSTM network, designed to handle and predict sequential patterns in time-series data. Unlike traditional LSTMs, BiLSTM consists of two independent LSTM units: one processes the input sequence in the forward direction (forward LSTM), and the other processes the sequence in the reverse direction (backward LSTM). This bidirectional structure allows BiLSTM to capture both forward and backward temporal features, adapt to delayed contamination effects, be more sensitive to anomalous fluctuations, and exhibit strong compatibility with other feature extraction modules. Therefore, the construction of the BiLSTM network is represented by Eqs. (9)–(11).

1. Forward LSTM Output $\vec{h}_t$:

$$\vec{h}_t = \text{LSTM}(\vec{h}_{t-1}, x_t) \quad (9)$$

2. Backward LSTM Output $\overleftarrow{h}_t$:

$$\overleftarrow{h}_t = \text{LSTM}(\overleftarrow{h}_{t+1}, x_t) \quad (10)$$

3. Bidirectional LSTM Output h_t :

$$h_t = [\vec{h}_t, \overleftarrow{h}_t] \quad (11)$$

Model construction

Simulated CVOCA can rapidly extract features from the entire AQI data process and connect with the BiLSTM model to build a simulated CVOCA-BiLSTM model for predicting AQI values. The process follows these steps:

Step 1: Construct Complex Input Data.

Convert the AQI time series into complex form, where:

Real part = original AQI values;

Imaginary part = the second-order differences + Fast Fourier Transform (FFT);

The second-order differences = the acceleration of AQI value;

FFT = amplitude + phase information.

Step 2: Feature Extraction Using Simulated CVOCA.

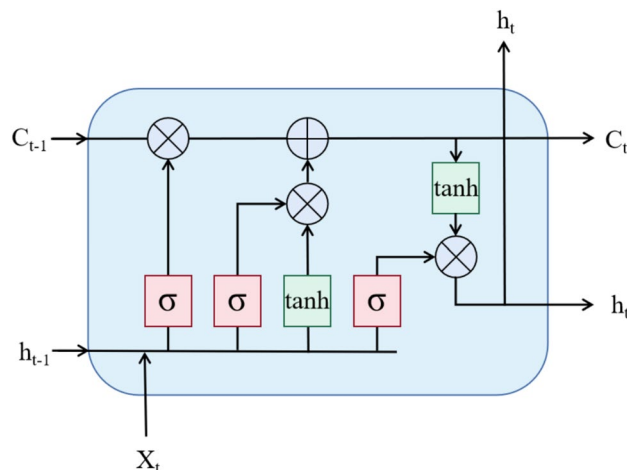


Fig. 2. LSTM architecture.

Perform sliding convolutions with multiple complex kernels on the input complex sequence to extract local trends and phase features.

Step 3: Connect to BiLSTM/LSTM Model.

Sliding one-dimensional convolutions employing multiple complex-valued kernels of length three were applied to the time series to extract local trend and phase features.

Step 4: Predict AQI Values.

Output the future AQI predictions through a fully connected layer.

The combination of BiLSTM and the complex features extracted by simulated CVOCA allows for deeper spatiotemporal feature fusion, enhancing the model's ability to learn both local trends and long-term dependencies.

In previous studies, hybrid models based on LSTM networks have been widely applied to AQI forecasting tasks^{22–24}. The innovation of this study lies in the introduction of the fast simulated CVOCA feature extraction combined with a three-layer BiLSTM model. The main distinction of this model from previous approaches is its ability to rapidly extract high-dimensional data features while capturing both forward and backward dependencies in time series data. Unlike LSTM, which is a unidirectional structure that processes data sequentially over time and is suitable for tasks where past information influences future predictions, BiLSTM, with its bidirectional structure, can simultaneously capture both past and future contextual information. This makes BiLSTM more effective in tasks that require a deeper understanding of time dependencies²⁵. BiLSTM is particularly well-suited for exploring the relationship of each data point in both forward and backward directions across the time dimension, which is especially crucial when making long-term predictions.

Figure 3 illustrates the structure of the simulated CVOCA-LSTM model, simplified to a sequence length of 1. As shown in the diagram, the 10-time-step input is sequentially fed into the first layer of the LSTM for training. The model's predicted output is then compared with the actual value at the future time point (t+1), and the loss is calculated. This process is repeated in the second and third layers of LSTM, with the cumulative loss from the three layers being used as the overall loss of the model. During training, the model is optimized until convergence. Finally, when the loss is minimized, the data is passed into the fully connected layer (Dense) to output the prediction.

Figure 4 illustrates the structure of the simulated CVOCA-BiLSTM model. Each layer contains two LSTM units: one for forward training and the other for backward training. For clarity, a sequence length of 1 is still used in this diagram. It is important to note that the data is simultaneously fed into both the forward and backward directions, and the loss is calculated separately for each direction. The results from both directions are then summed to compute the total loss for that layer. The output from the first layer of BiLSTM is concatenated and integrated before being passed into the second layer of BiLSTM, where the same process is repeated. This continues through the third layer, with the final output being processed by the fully connected layer to generate the prediction.

Model evaluation metrics

In this study, the performance of the proposed hybrid model is quantitatively assessed using five metrics: RMSE, MAE, MAPE, R, R^2 . Generally, lower values of RMSE, MAE, and MAPE, along with higher values of R and R^2 , indicate better model performance. The formulas for these evaluation metrics are presented in Eqs. (12)–(16).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2} \quad (12)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |x_i - \hat{x}_i| \quad (13)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{x_i - \hat{x}_i}{x_i} \right| \times 100\% \quad (14)$$

$$R = \frac{\sum_{i=1}^n \hat{x}_i x_i - \sum_{i=1}^n \hat{x}_i \sum_{i=1}^n x_i / n}{\sqrt{(\sum_{i=1}^n \hat{x}_i^2 - (\sum_{i=1}^n \hat{x}_i)^2 / n)(\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2 / n)}} \quad (15)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (x_i - \hat{x}_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (16)$$

Here, x_i represents the actual value at time step i , $\hat{x}_i$ denotes the predicted value at the corresponding time step, and $\bar{x}$ is the mean of all actual values x_i , where $i = 1, 2, \dots, n$, and n is the total number of samples. All data points from the test set are individually used to calculate the evaluation metrics, thereby assessing the predictive capability of the machine learning model for AQI forecasting.

Results and discussion

Following rapid feature extraction via simulated CVOCA, a CVOCA-BiLSTM model was constructed to predict AQI values. The performance of the proposed simulated CVOCA-BiLSTM model was validated using daily

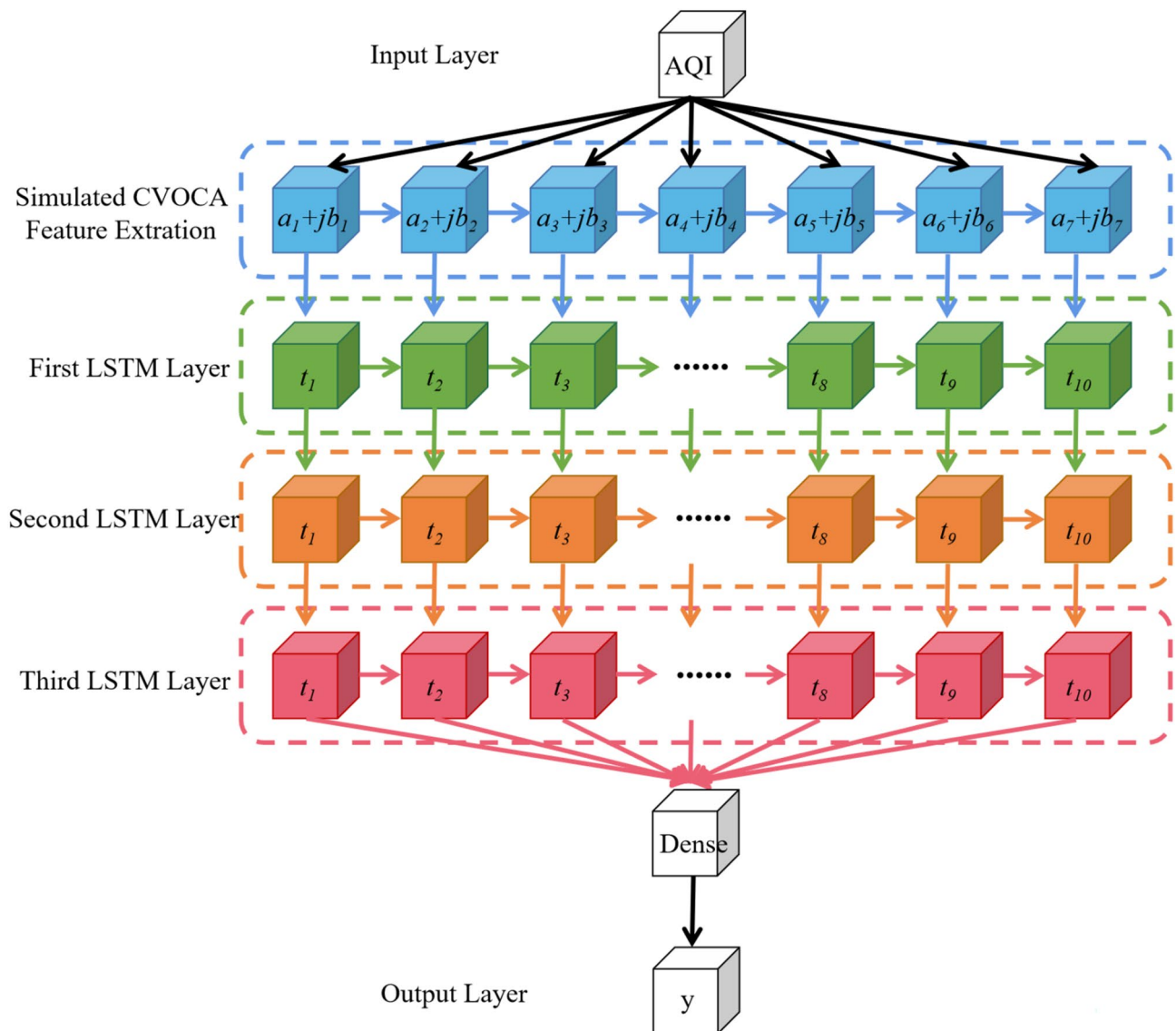


Fig. 3. Illustration of a three-layer simulated CVOCA-LSTM architecture for sequential data processing.

average AQI data from Lhasa. To demonstrate the advantages of the proposed approach, LSTM, BiLSTM, and simulated CVOCA-LSTM—were selected for comparative analysis. The prediction results and comparative evaluation of these models are discussed in detail in the following subsections.

Model optimization

Figure 5a illustrates the learning curves of the LSTM, BiLSTM, simulated CVOCA-LSTM, and simulated CVOCA-BiLSTM models, depicting changes in training and validation loss. It is evident that all four models adopt the Huber loss function, which integrates the advantages of both mean squared error (MSE) and mean absolute error (MAE). The Huber loss not only ensures continuity and differentiability of the loss function but also leverages the gradient behavior of MSE for small errors, allowing for more precise minimization. Furthermore, its robustness to outliers makes it particularly effective in handling anomalous data. Ultimately, the training and validation loss curves converge closely, indicating good generalization. Several regularization techniques were also employed to mitigate overfitting. For example, the early stopping mechanism was applied, halting training at around the 50th epoch out of 100 iterations to prevent overtraining. Dropout was used during training to randomly deactivate a portion of the neurons, forcing the model to learn under varied structures and enhancing generalization ability. Additionally, normalization and denormalization were employed—scaling features to a uniform range improved training speed, model stability, and activation function performance, while reducing gradient-related issues. Denormalizing the predictions allowed results to be restored to their original scale, making them easier to interpret, compare, and evaluate, and more suitable for practical applications.

As an adaptive study, initially, two-layer LSTM and BiLSTM models were used to train the system in order to minimize computational requirements. However, this configuration did not yield satisfactory accuracy

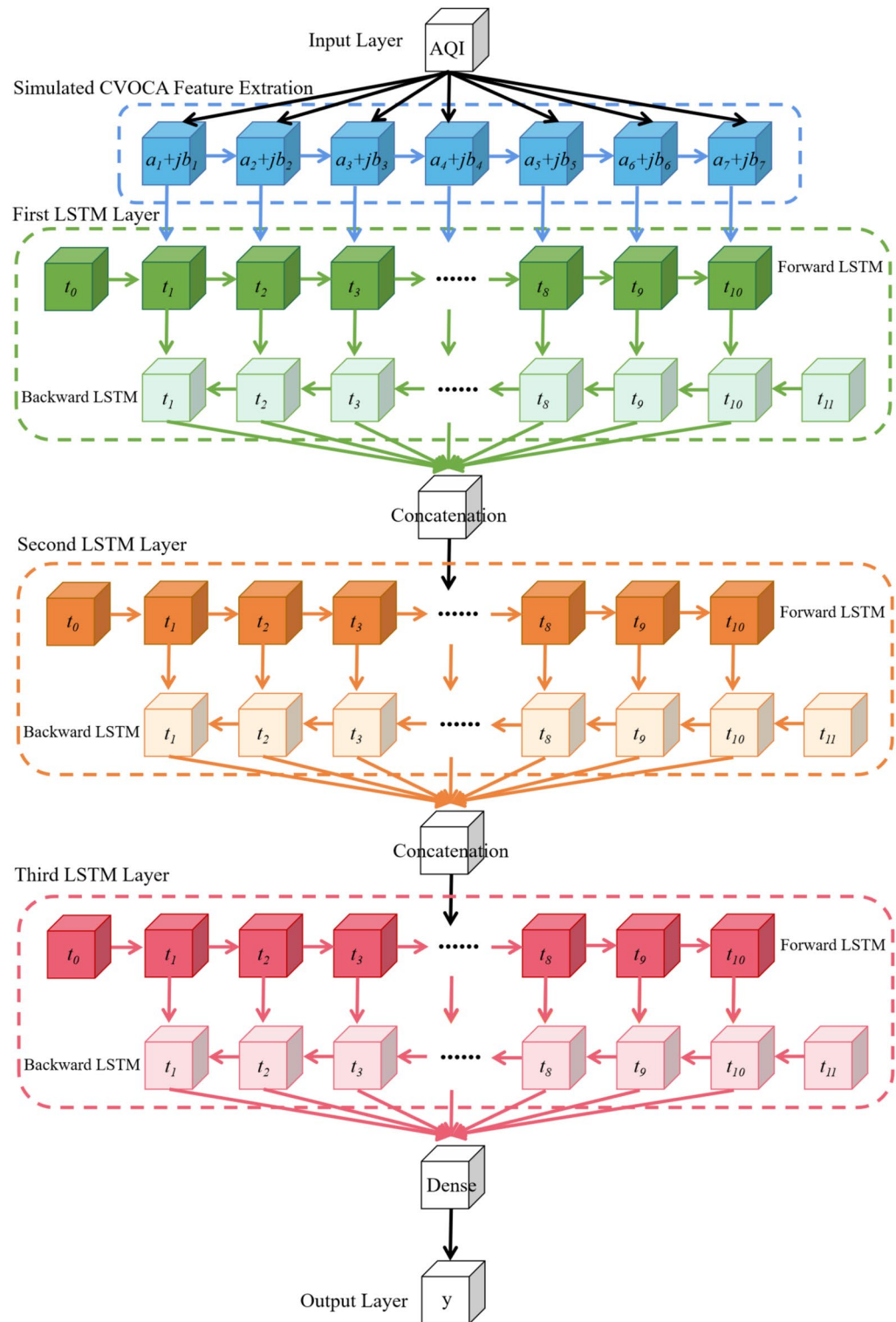


Fig. 4. Illustration of a three-layer simulated CVOCA-BiLSTM model architecture for sequential data processing.

results. Accordingly, feature extraction was performed using the CVOCA module, and the network depth was increased to three layers to optimize the model structure and performance. The real part was represented by the original values, while the second-order differences (acceleration) and FFT were used as the imaginary part to represent frequency domain features. The second-order difference was chosen for its ability to enrich the feature space with higher-order temporal dynamics, improving sensitivity to local variations that may be missed with first-order changes. Experimentally, models using the second-order difference achieved lower prediction

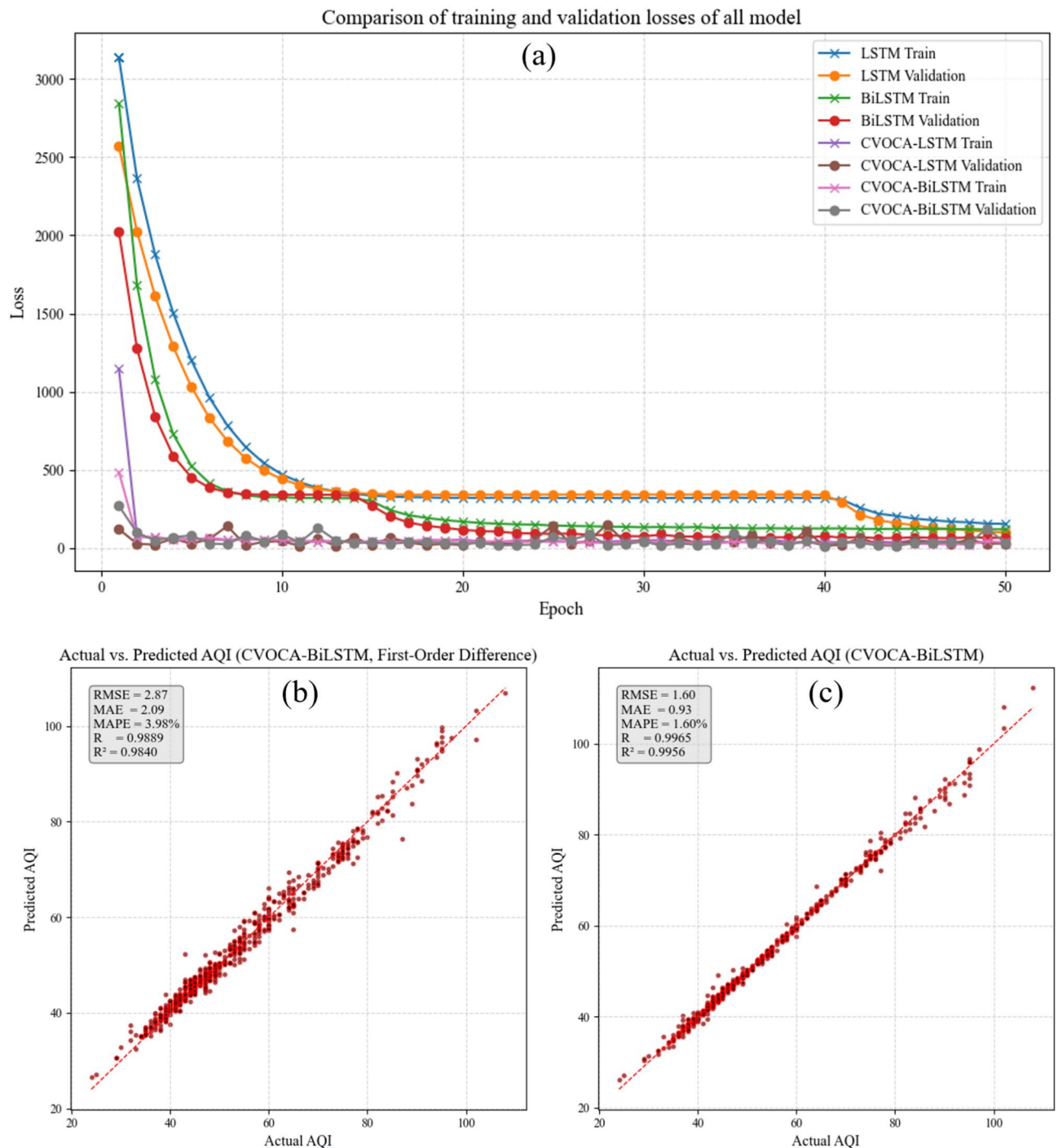


Fig. 5. The training loss and validation loss of the Huber function of the four models (a), Comparison of models based on first-order (b) and second-order differences (c).

errors (e.g., reduced RMSE and MAPE) and better captured inflection points in the AQI trajectory (Fig. 5b, c). This complementary information enhanced the model's discriminative power and robustness. This approach elevated the original time series data into a more complex feature space, enhancing the predictive capability, which resulted in outcomes closer to the expected results.

Prediction performance

The comparison of observed and predicted AQI series for Lhasa, Shanghai, and Beijing (Fig. 6) showed that the simulated CVOCA-BiLSTM hybrid model achieved the highest predictive accuracy across environments, reproducing baseline trends while preserving short-term fluctuations. In high-altitude, low-variability Lhasa, the hybrid model produced the narrowest residuals centered near zero, indicating that amplitude-phase

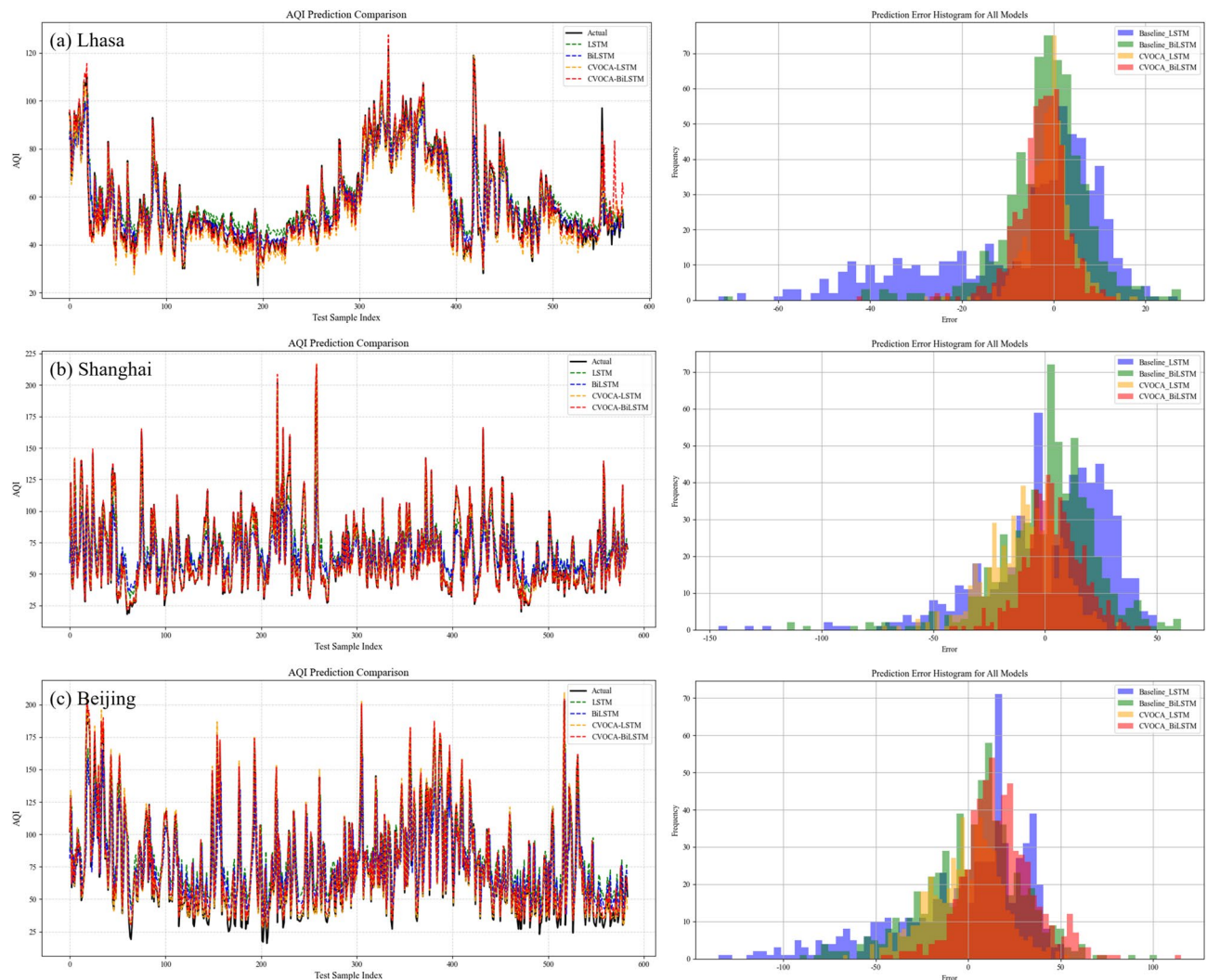


Fig. 6. Comparison of actual AQI values in Lhasa (a), Shanghai (b), and Beijing (c) with predictions from four models, including error histograms.

feature enhancement and bidirectional temporal fusion preserved subtle signals and reduced systematic bias. In coastal Shanghai, it more faithfully tracked transient oscillations but exhibited greater residual variance owing to stronger local and regional variability. In highly variable Beijing, overall trend agreement remained strong, though residual tails were heavier, reflecting sensitivity to sporadic, externally driven extremes. Model comparisons indicated that simulated CVOCA mitigated non-stationarity and improved peak reconstruction, while BiLSTM reduced information loss via forward–backward fusion; together they yielded robust cross-scenario performance. In summary, the simulated CVOCA-BiLSTM model demonstrates remarkable strengths in predictive accuracy, environmental adaptability, and methodological innovation.

Model evaluation

Figure 7 presents scatter plots comparing the predicted and actual AQI values for each model, showing a progressive improvement in model performance. The proposed simulated CVOCA-BiLSTM model achieves the highest coefficient of determination, with R and R^2 values of 0.997 and 0.996, respectively—significantly outperforming both LSTM and BiLSTM. This superior performance is attributed to the feature enhancement mechanism of simulated CVOCA, which markedly boosts the model's predictive capabilities. Additionally, it is evident that the models incorporating bidirectional temporal structures, such as simulated CVOCA-BiLSTM and BiLSTM, demonstrate slightly better performance than their unidirectional counterparts. This indicates that bidirectional structures further enhance the extraction of temporal dependencies. In terms of MAPE, the standard LSTM exceeds 10%, indicating relatively large prediction errors at certain points—suggesting it may be less reliable in practical applications. In contrast, both simulated CVOCA-BiLSTM and simulated CVOCA-LSTM models achieve MAPE values below 5%, demonstrating excellent control over relative errors. The simulated CVOCA-BiLSTM model also records the lowest RMSE and MAE values, at 1.60 and 0.93, respectively—highlighting its superior overall accuracy.

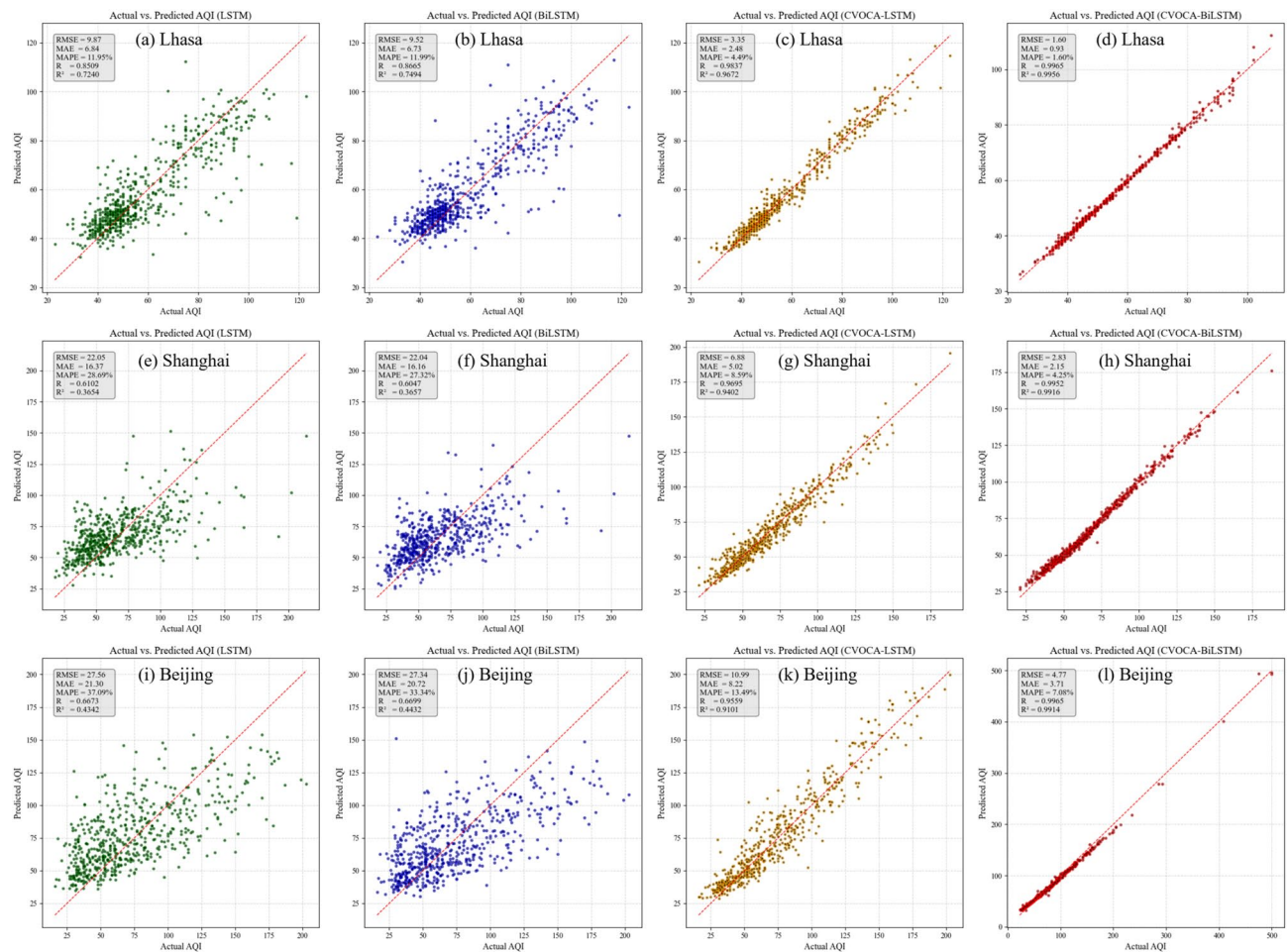


Fig. 7. Scatter plots comparing predicted and actual AQI values from the four models for Lhasa (a–d), Shanghai (e–h), and Beijing (i–l).

Comparative analysis

Robustness and generalizability

Across evaluations in Lhasa, Shanghai, and Beijing, the simulated CVOCA-BiLSTM consistently achieved the highest AQI prediction accuracy, with R^2 values of 0.996, 0.992, and 0.991, respectively, and corresponding RMSE of 1.60, 2.83, and 4.77 (Table 1; Fig. 7). These results demonstrate clear advantages over LSTM, BiLSTM, and simulated CVOCA-LSTM, particularly in capturing high-frequency fluctuations and reducing local deviations. Compared with recent approaches such as BWO-CAformer ($R^2=0.931$; RMSE=9.82) and pollutant-level process frameworks (e.g., GML), the hybrid model showed superior accuracy without relying on decomposition or preprocessing strategies. By integrating simulated CVOCA's amplitude-phase feature enhancement with bidirectional sequence modeling, the proposed method robustly preserved both baseline levels and transient oscillations, achieving stable and generalizable forecasting across plateau and plain-city environments.

Computational efficiency and scalability

As shown in Table 2, the integration of the CVOCA module markedly enhances both the computational efficiency and scalability of LSTM and BiLSTM models. Specifically, CVOCA-enhanced models achieve approximately $2.5 \times$ faster inference and substantially reduced training times per epoch compared to their vanilla counterparts, while maintaining moderate memory usage (< 1 GB). The inference time per sample of the CVOCA-LSTM is roughly one-third that of the baseline LSTM, and the simulated CVOCA-BiLSTM reaches an average prediction time of 0.00049 s per sample, representing a 59.3% speedup over the standard BiLSTM. Importantly, despite modest increases in total trainable parameters, both CVOCA-LSTM and CVOCA-BiLSTM exhibit significantly improved predictive performance ($R^2=0.967$ and 0.996, respectively), demonstrating that the architecture effectively leverages the additional capacity without compromising efficiency. These quantitative results provide solid evidence for the high computational efficiency and scalability of the proposed framework, underscoring its potential for large-scale, high-dimensional AQI forecasting tasks.

Subject	Method	Region	Time	Best result	References
AQI	Simulated CVOCA-BiLSTM	Lhasa, China	2025	$R^2 = 0.996$ $R = 0.997$ $MAE = 0.93$ $MAPE = 1.60\%$ $RMSE = 1.60$	This study
AQI	Simulated CVOCA-BiLSTM	Shanghai, China	2025	$R^2 = 0.992$ $R = 0.995$ $MAE = 2.15$ $MAPE = 4.25\%$ $RMSE = 2.83$	This study
AQI	Simulated CVOCA-BiLSTM	Beijing, China	2025	$R^2 = 0.991$ $R = 0.997$ $MAE = 3.71$ $MAPE = 7.08\%$ $RMSE = 4.77$	This study
AQI	BWO-CAformer	Beijing/ Wuhan, China	2025	$R^2 = 0.931$ $MAE = 6.909$ $MAPE = 15.265\%$ $RMSE = 9.822$	9
AQI	LR/DTR/RF/KNN/KNR	Beijing/ Shanghai/Hubei/ Shaanxi, China	2025	$R^2 = 0.9995$ (RF) $RMSE = 0.8660$ (RF)	26
AQI	GRU/CNN/FNN/LSTM/MLP	Azamgarh, India	2025	$R^2 = 0.9971$ (FNN) $MAE = 2.89$ (FNN) $RMSE = 4.99$ (FNN)	27
AQI	TimeGAN	Beijing/ Tangshan, China	2025	$MAE = 3.651$ $MAPE = 5.072\%$ $RMSE = 6.407$	28
AQI	STL-Metis-MHBA-TC	Beijing/ Wuxi/ Guangzhou, China	2025	$R^2 = 0.9971$ $MAE = 2.1744$ $MAPE = 4.96\%$ $RMSE = 3.3636$	29
AQI	RF/XGBoost/LightGBM/CatBoost/GB/KNR	Chhattisgarh, India	2025	$R^2 = 0.93$ $MAE = 0.20$ $MSE = 0.08$ $RMSE = 0.29$	30
AQI	LR/RF/DTR	Delhi, India	2025	$R^2 = 0.99$ (RF) $RMSE = 0.27$ (RF)	31
AQI	XGBoost/LarsCV/Bayesian Ridge/AdaBoost	Hyderabad, India	2025	$R^2 = 0.999$ $MAE = 0.496$ $MSE = 0.429$ $RMSE = 0.655$	16
PM _{2.5}	MLP/SVR/RF	Beijing, China /Istanbul, Turkey	2025	$R^2 = 0.91$ $MAE = 6.67$ $RMSE = 8.80$	32
PM _{2.5}	OA-LSTM	Tehran, Iran	2025	$R^2 = 0.88$ $RMSE = 3.01$	33
PM _{2.5}	CNN/ANN	Hefei, China	2025	$R^2 = 0.88$ (CNN) $R^2 = 0.74$ (ANN)	34
PM _{2.5}	CNN-Transformer	Cangzhou, China	2025	$R^2 = 0.930$ $MAE = 9.504$ $MAPE = 18.097\%$ $RMSE = 13.026$	35
PM ₁₀	RF/KNN/WKNN/SVR/LASSO	Zabol, Iran	2025	$R^2 = 0.75$ (RF) $MAE = 244.87$	36
NO ₂	ANFIS-FCM/ ANN/ARMA/CNN/GRU/LSTM/CNN-GRU/CNN-LSTM	Istanbul/Ankara, Turkey	2025	$R^2 = 0.9547$ $MSE = 0.474$ $MAE = 0.252$	12
NO ₂	SLNet	England, UK	2025	$R^2 = 0.914$ $MAE = 3.405$ $RMSE = 4.898$	37
O ₃	LSTM/ANN	Liaocheng, China	2025	$R^2 = 0.689$ $MAE = 21.384$ $RMSE = 27.466$	38
PM _{2.5} /O ₃	GBDT/XGBoost/RF	Wuhan, China	2025	$R^2 = 0.987$ (PM _{2.5}) $MAE = 3.909$ (PM _{2.5}) $RMSE = 6.784$ (PM _{2.5}) $R^2 = 0.947$ (O ₃) $MAE = 3.942$ (O ₃) $RMSE = 6.323$ (O ₃)	39
Continued					

Subject	Method	Region	Time	Best result	References
PM _{2.5} / PM ₁₀ / O ₃	PSO-LSTM	Harbin, China	2025	R ² = 0.9486 (PM _{2.5}) MAE = 0.0014 (PM _{2.5}) RMSE = 0.0033 (PM _{2.5}) R ² = 0.9419 (PM ₁₀) MAE = 0.0037 (PM ₁₀) RMSE = 0.0077 (PM ₁₀) R ² = 0.9698 (O ₃) MAE = 0.0113 (O ₃) RMSE = 0.0184 (O ₃)	40
PM _{2.5} / PM ₁₀ / O ₃ / NO ₂ / SO ₂ / CO	GML	Global	2025	R ² = 0.97 (PM _{2.5}) RMSE = 9.43 (PM _{2.5}) R ² = 0.93 (PM ₁₀) RMSE = 8.99 (PM ₁₀) R ² = 0.95 (O ₃) RMSE = 17.37 (O ₃) R ² = 0.97 (NO ₂) RMSE = 2.88 (NO ₂) R ² = 0.80 (SO ₂) RMSE = 2.55 (SO ₂) R ² = 0.96 (CO) RMSE = 3.50 (CO)	41

Table 1. Comparative summary of recent AQI/pollutant forecasting models and their best-reported performance across regions.

Model	Total trainable parameters	Training time/epoch (s)	Total training time (s)	Inference time/sample (ms)	Peak RAM (GB)	Predictive Performance Metrics
LSTM (baseline)	1,350,913	6.90	689.99	1.2587	0.598	RMSE = 9.87 MSE = 6.84 MAPE = 11.95% R = 0.8509 R ² = 0.7240
BiLSTM (baseline)	3,750,145	9.74	974.10	1.2076	0.777	RMSE = 9.52 MSE = 6.73 MAPE = 11.99% R = 0.8665 R ² = 0.7494
Simulated CVOCA-LSTM	1,361,153	4.21	421.03	0.4687	0.635	RMSE = 3.35 MSE = 2.48 MAPE = 4.49% R = 0.9837 R ² = 0.9672
Simulated CVOCA-BiLSTM	3,881,601	5.06	506.49	0.4913	0.820	RMSE = 1.60 MSE = 0.93 MAPE = 1.60% R = 0.9965 R ² = 0.9956

Table 2. Compare and analyze the quantitative data on the computational efficiency and cost of the four models.

Prediction of AQI values

Using the LSTM, BiLSTM, simulated CVOCA-LSTM, and simulated CVOCA-BiLSTM models developed in this study, we forecasted AQI levels for January 2025 in three representative cities—Lhasa, Shanghai, and Beijing. The results are shown in Fig. 8, where distinct prediction patterns emerged across these cities, reflecting their contrasting environmental contexts as well as the evolution and driving mechanisms of eco-environmental quality across different urbanization stages⁴².

In Lhasa, where AQI time series are relatively smooth due to limited local emission sources and high-altitude boundary-layer dynamics, the simulated CVOCA-BiLSTM produced a stable prediction curve that closely aligned with historical variability. Specifically, for January 2025, the average AQI predicted by simulated CVOCA-BiLSTM was stable at 52.63, compared to the actual average AQI value of 53 for January 2014–2024. This makes simulated CVOCA-BiLSTM’s prediction more accurate than the other three models for January 2025, leading to the conclusion that the AQI level for the first month of 2025 falls within the “Good” category (Table 3). In contrast, the LSTM model predicted an average AQI of 45.56, indicating an “Excellent” quality level^{43,44}, which significantly differed from the other models. While the prediction trends for pollution levels from simulated CVOCA-LSTM and simulated CVOCA-BiLSTM were similar in terms of pattern changes, the AQI levels showed some variations. This could be attributed to the bidirectional time series structure, which allows for a more accurate overall prediction of AQI values over time. The differences in AQI levels predicted by the LSTM, BiLSTM, simulated CVOCA-LSTM, and simulated CVOCA-BiLSTM models explain the variations in model accuracy and precision. In contrast, the prediction curves for Shanghai and Beijing exhibited more

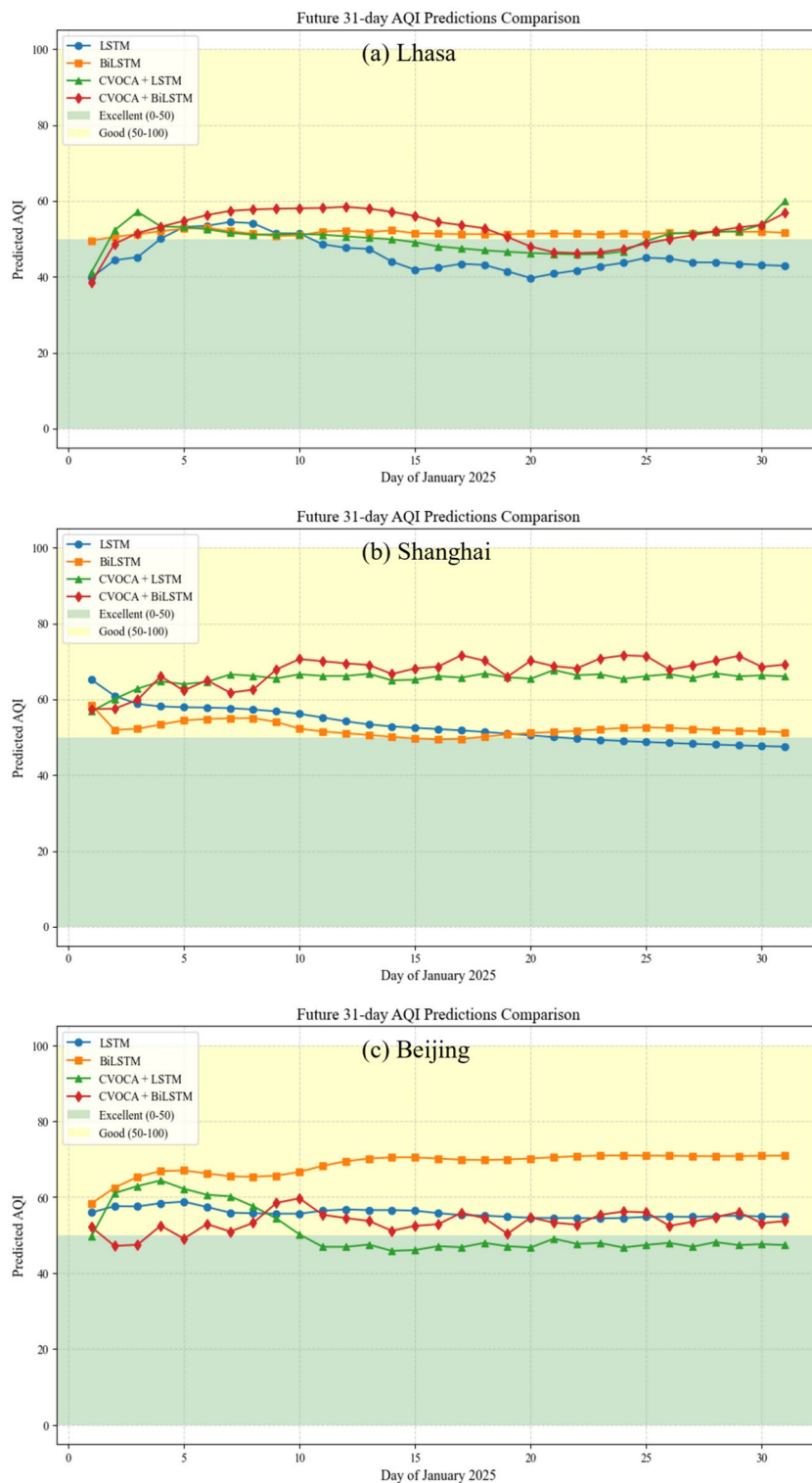


Fig. 8. Predictions of the AQI for January 2025 in Lhasa (a), Shanghai (b), and Beijing (c) by four models.

pronounced short-term fluctuations, consistent with their denser emission sources and stronger meteorological variability. Across both plain cities, the simulated CVOCA-BiLSTM captured these fluctuations more effectively than the other models, providing forecasts that preserved both baseline levels and transient variations. Notably, the conventional LSTM tended to underestimate AQI in all three cities, with particularly smoothed predictions in Lhasa, failing to capture the subtle dynamics of plateau environments. BiLSTM improved upon this by partially incorporating temporal dependencies, while simulated CVOCA-LSTM benefited from enhanced feature extraction but still exhibited limitations in reproducing bidirectional temporal dynamics. By contrast,

AQI	Grade	AQI classes	Notes
0–50	I	Excellent	The air quality is very good, with little impact on health
51–100	II	Good	May have a slight impact on a small number of sensitive individuals
101–150	III	Light pollution	It has some effect on sensitive people
151–200	IV	Moderate pollution	It affects most people
201–300	V	Serious pollution	It affects everyone
> 300	VI	Heavy pollution	Will have a significant impact on the health of all populations

Table 3. The standard Air Quality Index and related information, for China.

the simulated CVOCA-BiLSTM consistently yielded the most balanced forecasts, simultaneously maintaining stability in plateau conditions and adaptability to the higher variability observed in plain cities.

These results highlight that the hybrid model not only excels under the relatively low-fluctuation conditions of high-altitude cities but also generalizes effectively to plain urban environments characterized by more dynamic AQI patterns. The differences in AQI levels predicted by the LSTM, BiLSTM, simulated CVOCA-LSTM, and simulated CVOCA-BiLSTM models explain the variations in model accuracy and precision.

Conclusions

This study proposes a novel simulated CVOCA-BiLSTM hybrid model for high-accuracy AQI prediction in the high-altitude environment of Lhasa, Tibet. The model integrates the ultra-fast feature extraction capability of the simulated CVOCA for high-dimensional AQI data and the bidirectional time series dependency modeling ability of the BiLSTM network. Experimental results demonstrate that the proposed model achieves outstanding predictive performance, with RMSE, MAE, MAPE, R, and R² values of 1.60, 0.93, 1.60%, 0.997, and 0.996, respectively, outperforming standalone LSTM, BiLSTM, and conventional hybrid models. The synergy between the simulated CVOCA's capacity to process complex amplitude–phase features and BiLSTM's bidirectional learning capability effectively addresses the nonlinear dynamics, sudden anomalies, and pollution lag effects in AQI time series. Validation using actual data from Lhasa (2014–2024) confirms the model's robustness under the unique high-altitude setting, while additional assessments in Shanghai (coastal plain with moderate variability) and Beijing (inland plain with high variability) demonstrate its transferability across contrasting geographical and variability regimes. This research framework not only enhances the accuracy of air quality predictions but also contributes to data-driven urban environmental management, offering feasible decision support for public health protection in ecologically fragile areas. Future research will extend to multi-pollutant coupling mechanisms and regional-scale spatiotemporal dynamics to further improve the model's general applicability.

Data availability

The data used in this study were obtained from the China National Environmental Monitoring centre (<https://www.cnemc.cn/>) and historical data from the AQI Study database (<https://www.aqistudy.cn/historydata/>).

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References

- Gewin, V. How a dangerous stew of air pollution is choking the United States. *Nature* **612**, 204–207. <https://doi.org/10.1038/d41586-022-04333-9> (2022).
- Canadell, J. G. et al. Multi-decadal increase of forest burned area in Australia is linked to climate change. *Nat. Commun.* **12**(1), 6921. <https://doi.org/10.1038/s41467-021-27225-4> (2021).
- Balsari, S. & Mohanan, M. Local government actions can curb air pollution in India and Pakistan. *Nature* **634**, 1055–1056. <https://doi.org/10.1038/d41586-024-03314-4> (2024).
- Li, X. et al. Polar regions are critical in achieving global sustainable development goals. *Nat. Commun.* **16**(1), 3879. <https://doi.org/10.1038/s41467-025-59178-3> (2025).
- Zhao, S., Feng, T., Tie, X. & Wang, Z. The warming Tibetan Plateau improves winter air quality in the Sichuan Basin. *China. Atmos. Chem. Phys.* **20**(23), 14873–14887. <https://doi.org/10.5194/acp-20-14873-2020> (2020).
- Yin, X. et al. Gaseous and particulate pollutants in Lhasa, Tibet during 2013–2017: Spatial variability, temporal variations and implications. *Environ. Pollut.* **253**, 68–77. <https://doi.org/10.1016/j.envpol.2019.06.113> (2019).
- Liang, Q. et al. Spatial–temporal characteristics and influencing mechanisms of air quality index by considering COVID-19 in Yunnan. *Southeastern Tibetan Plateau. Atmosphere* **14**(2), 378. <https://doi.org/10.3390/atmos14020378> (2023).
- Sokhi, R. S. et al. Advances in air quality research—current and emerging challenges. *Atmos. Chem. Phys.* **22**(7), 4615–4703. <https://doi.org/10.5194/acp-22-4615-2022> (2022).
- Dong, X., Li, D., Wang, W. & Shen, Y. BWO-CAformer: An improved Informer model for AQI prediction in Beijing and Wuhan. *Process Saf. Environ. Prot.* **195**, 106800. <https://doi.org/10.1016/j.psep.2025.106800> (2025).
- Sarwar, G. et al. Representing particulate nitrate photolysis over seawater improves CMAQ ozone predictions over the contiguous United States. *Sci. Total Environ.* **970**, 178968. <https://doi.org/10.1016/j.scitotenv.2025.178968> (2025).
- Zhang, H. et al. Contribution of dust emissions from farmland to particulate matter concentrations in North China Plain: Integration of WRF-Chem and WEPS model. *Environ. Int.* **195**, 109191. <https://doi.org/10.1016/j.envint.2024.109191> (2025).
- Uluocak, I., Pinar, E. & Bilgili, M. Atmospheric NO₂ concentration prediction with statistical and hybrid deep learning methods. *Environ. Ecol. Stat.* **32**, 89–118. <https://doi.org/10.1007/s10651-024-00637-3> (2025).
- Wang, S., Ren, Y. & Xia, B. Estimation of urban AQI based on interpretable machine learning. *Environ. Sci. Pollut. Res.* **30**(42), 96562–96574. <https://doi.org/10.1007/s11356-023-29336-5> (2023).

14. Mu, X., Wang, S., Jiang, P. & Wu, Y. Estimation of surface ozone concentration over Jiangsu province using a high-performance deep learning model. *J. Environ. Sci.* **132**, 122–133. <https://doi.org/10.1016/j.jes.2022.09.032> (2023).
15. Ahmed, D. M., Hassan, M. M. & Mstafa, R. J. A review on deep sequential models for forecasting time series data. *Appl. Comput. Intell. Soft Comput.* <https://doi.org/10.1155/2022/6596397> (2022).
16. Ravindiran, G. et al. Ensemble stacking of machine learning models for air quality prediction for Hyderabad City in India. *iScience* **28**(2), 111894. <https://doi.org/10.1016/j.isci.2025.111894> (2025).
17. Wu, Q. & Lin, H. Daily urban air quality index forecasting based on variational mode decomposition, sample entropy and LSTM neural network. *Sustain. Cities Soc.* **50**, 101657. <https://doi.org/10.1016/j.scs.2019.101657> (2019).
18. Dong, J., Zhang, Y. & Hu, J. Short-term air quality prediction based on EMD-transformer-BiLSTM. *Sci. Rep.* **14**(1), 20513. <https://doi.org/10.1038/s41598-024-67626-1> (2024).
19. Bai, Y. et al. TOPS-speed complex-valued convolutional accelerator for feature extraction and inference. *Nat. Commun.* **16**(1), 292. <https://doi.org/10.1038/s41467-024-55321-8> (2025).
20. Hochreiter, S. & Schmidhuber, J. Long short-term memory. *Neural Comput.* **9**(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735> (1997).
21. Graves, A. & Schmidhuber, J. Framewise phoneme classification with bidirectional LSTM and other neural network architectures. *Neural Netw.* **18**(5–6), 602–610. <https://doi.org/10.1016/j.neunet.2005.06.042> (2025).
22. Zhang, J. & Li, S. Air quality index forecast in Beijing based on CNN-LSTM multi-model. *Chemosphere* **308**, 136180. <https://doi.org/10.1016/j.chemosphere.2022.136180> (2022).
23. Rabie, R., Asghari, M., Nosrati, H., Niri, M. E. & Karimi, S. Spatially resolved air quality index prediction in megacities with a CNN-Bi-LSTM hybrid framework. *Sustain. Cities Soc.* **109**, 105537. <https://doi.org/10.1016/j.scs.2024.105537> (2024).
24. Wang, J. et al. An air quality index prediction model based on CNN-ILSTM. *Sci. Rep.* **12**(1), 8373. <https://doi.org/10.1038/s41598-022-12355-6> (2022).
25. Fan, Y., Tang, Q. & Guo, Y. BiLSTM-MLAM: a multi-scale time series prediction model based on Bi-LSTM and local attention mechanism. <https://doi.org/10.21203/rs.3.rs-4229520/v1> (2024).
26. Mubeen, M. et al. Smart prediction and optimization of air quality index with artificial intelligence. *J. Environ. Sci.* <https://doi.org/10.1016/j.jes.2025.02.041> (2025).
27. Ansari, A. & Quaff, A. R. Data-driven analysis and predictive modelling of hourly air quality index (AQI) using deep learning techniques: A case study of Azamgarh, India. *Theor. Appl. Climatol.* **156**(1), 74. <https://doi.org/10.1007/s00704-024-05304-y> (2025).
28. Wang, K., Yang, T., Kong, S. & Li, M. Air quality index prediction through TimeGAN data recovery and PSO-optimized VMD-deep learning framework. *Appl. Soft Comput.* **170**, 112626. <https://doi.org/10.1016/j.asoc.2024.112626> (2025).
29. Qian, S. et al. A novel ensemble framework based on intelligent weight optimization and multi-model fusion for air quality index prediction. *Urban Clim.* **59**, 102233. <https://doi.org/10.1016/j.uclim.2024.102233> (2025).
30. Gond, A. K., Jamal, A. & Verma, T. Developing a machine learning model using satellite data to predict the air quality index (AQI) over Korba Coalfield, Chhattisgarh (India). *Atmos. Pollut. Res.* **16**(2), 102398. <https://doi.org/10.1016/j.apr.2024.102398> (2025).
31. Pande, C. B. et al. Forecasting of monthly air quality index and understanding the air pollution in the urban city, India based on machine learning models and cross-validation. *J. Atmos. Chem.* **82**(1), 1. <https://doi.org/10.1007/s10874-024-09466-x> (2025).
32. Emeç, M. U. R. A. T. & Yurtsever, M. A novel ensemble machine learning method for accurate air quality prediction. *Int. J. Environ. Sci. Technol.* **22**(1), 459–476. <https://doi.org/10.1007/s13762-024-05671-z> (2025).
33. Kaveh, M., Mesgari, M. S. & Kaveh, M. A novel evolutionary deep learning approach for PM_{2.5} prediction using remote sensing and spatial-temporal data: A case study of Tehran. *ISPRS Int. J. Geo Inf.* **14**(2), 42. <https://doi.org/10.3390/ijgi14020042> (2025).
34. Xia, S., Zhang, R., Zhang, L., Wang, T. & Wang, W. Multi-dimensional distribution prediction of PM_{2.5} concentration in urban residential areas based on CNN. *Build. Environ.* **267**, 112167. <https://doi.org/10.1016/j.buildenv.2024.112167> (2025).
35. Wang, Y. Z., He, H. D., Huang, H. C., Yang, J. M. & Peng, Z. R. High-resolution spatiotemporal prediction of PM_{2.5} concentration based on mobile monitoring and deep learning. *Environ. Pollut. Res.* **364**, 125342. <https://doi.org/10.1016/j.envpol.2024.125342> (2025).
36. Kalantari, E., Gholami, H., Malakooti, H., Kaskaoutis, D. G. & Saneei, P. An integrated feature selection and machine learning framework for PM₁₀ concentration prediction. *Atmos. Pollut. Res.* **16**(5), 102456. <https://doi.org/10.1016/j.apr.2025.102456> (2025).
37. Wang, S., Zhang, S., Wang, D. & Li, W. Spatial-and-local-aware deep learning approach for Ground-Level NO₂ estimation in England with multisource data from satellite-based observations and chemical transport models. *Int. J. Appl. Earth Obs. Geoinf.* **139**, 104506. <https://doi.org/10.1016/j.jag.2025.104506> (2025).
38. Guo, Q., He, Z. & Wang, Z. Assessing the effectiveness of long short-term memory and artificial neural network in predicting daily ozone concentrations in Liaocheng City. *Sci. Rep.* **15**(1), 6798. <https://doi.org/10.1038/s41598-025-91329-w> (2025).
39. Xu, J., Liu, M., Chao, Y. & Chen, H. A novel prediction framework for estimating high spatial resolution near-ground PM_{2.5} and O₃ concentrations at street-level in urban areas. *Build. Environ.* **267**, 112141. <https://doi.org/10.1016/j.buildenv.2024.112141> (2025).
40. Chen, M. et al. Air pollution prediction based on optimized deep learning neural networks: PSO-LSTM. *Atmos. Pollut. Res.* **16**(3), 102413. <https://doi.org/10.1016/j.apr.2025.102413> (2025).
41. Anggraini, T. S., Irie, H., Sakti, A. D. & Wikantika, K. Global Air Quality Index Prediction using integrated spatial observation data and geographics machine learning. *Sci. Remote Sens.* **11**, 100197. <https://doi.org/10.1016/j.srs.2025.100197> (2025).
42. He, Z., Guo, Q., Yin, Z., Luo, X. & Sun, M. Evolution and driving mechanisms of eco-environmental quality across different urbanization stages. *Sci. Rep.* **15**, 22101. <https://doi.org/10.1038/s41598-025-06084-9> (2025).
43. Wang, Z. et al. Prospect of China's ambient air quality standards. *J. Environ. Sci.* **123**, 255–269. <https://doi.org/10.1016/j.jes.2022.03.036> (2023).
44. Guo, Q., He, Z. & Wang, Z. Monthly climate prediction using deep convolutional neural network and long short-term memory. *Sci. Rep.* **14**, 17748. <https://doi.org/10.1038/s41598-024-68906-6> (2024).

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Author contributions

Data curation, F.X.; methodology, F.X.; writing—original draft preparation, F.X.; formal analysis, X.C.; supervision, G.J. and D.B.; writing—review and editing, F.X. and Q.Z. All the authors have read and agreed to the published version of the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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